

Challenge: Measure the needed dimensions of the object provided, knowing that you will be recreating this object as a dimensioned solid model in a related homework assignment.

Student Learning Outcomes:

- 1) Develop ability to use dimensional measurement tools (instruments) rulers, calipers and micrometers
- 2) Identify resolution of each dimensional tool
- 3) Report percent differences and standard deviations for a population of measured data

Pre-lab:

- Familiarize yourself with how to use a standard ruler, a caliper, and a micrometer.
 - In iLearn you will find practice guides, user manuals, and/or video links
 - You may also use web based open resources to aid your work
- You must provide information on resources (those provided and ones you locate), using proper citation, when writing up your Challenge Report.
- If you are NOT using a resource, because you have prior experience with an instrument, provide discussion of your background expertise.
- Be prepared to write a brief description of the process steps you use to conduct the measurements.
- Zero the digital instruments before taking measurements. See the equipment manuals or ask the TA for help.

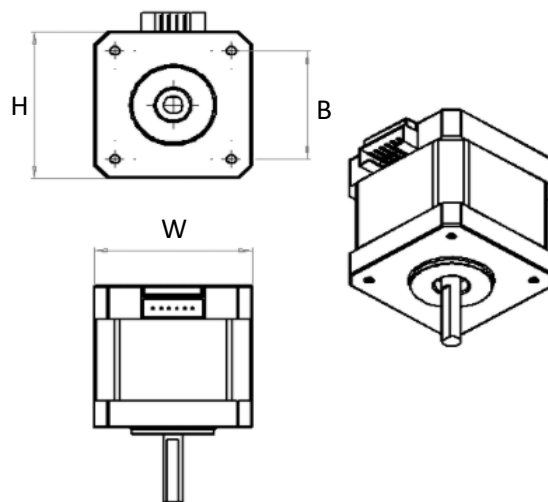


Figure 1 - NEMA Stepper Dimensions

Lab Guidance:

- Measure the width (W), and height (H), bolt spacing (B) dimensions of the object as shown in Figure 1. Choose two additional features on the object to measure for a total of five separate measured features on the object.
- Collect 5 separate measurements of each feature with two different instrument types for a total of 10 readings per feature. Available instruments include digital caliper, Vernier caliper, ruler, digital micrometer, Vernier micrometer.
- Prepare an Excel spreadsheet containing all measurements from both instruments for each dimension and instruments (ruler and Vernier caliper OR ruler and digital caliper). Be sure to indicate UNITS!

For example:

Object Dimension	Instrument 1	Instrument 2
	Reading 1	Reading 1
	...	...
	Reading 5	Reading 5

- To randomize the test, alternate use of the instruments (read dimension 1 using instrument 1, THEN read dimension 1 using instrument 2, etc)

Data Analysis:

- In the spreadsheet, calculate the mean and the standard deviation for each dimension from the five readings entered for each instrument.
- Determine the percent difference between readings on a given dimension using the two instruments.

Deliverables:

Upload the Challenge Report by the deadline, uploading to iLearn as two files

(a) Fully annotated spreadsheet file:

- Provide a spreadsheet file in Excel to document your raw data and your data analysis
- Be sure to use text blocks in the Excel file to annotate your work
- For example, if you are using a formula, show the formula

(b) Working Notes (PDF).

- Document your procedures in a Working Notes file, uploaded as a PDF.
- Show photos and/or sketches of your work
- Provide written notes of how you did the work
- Include observations of your work such as if you had difficulty with something, what did you do to resolve the issue, what kinds of troubleshooting did you undertake, etc
- Provide a summary of your data analysis
- Provide conclusions you make based on the data